

View Review

Paper ID

707

Paper Title

An Efficient Supervised Machine Learning Models for Fraud Transaction Detection: A Big Data Perspective in Finance

Track Name

Track 5: Machine Learning, Autonomous Systems, and Robotics

REVIEW QUESTIONS**1. Is the paper within the scope of the conference?**

Yes

2. Novelty of the paper

High

3. Are the presented concepts well-articulated, and are the contributions effectively highlighted?

Yes

4. Is the presentation well-structured in terms of language clarity, figure and table quality, and overall organization?

Excellent

5. Does the manuscript include a comprehensive literature review with properly cited references?

Excellent

6. Overall Evaluation:

Strong Accept

7. Confidence of the reviewer in the topic of the manuscript

High

8. Reviewer Comments to the Authors

The article An Efficient Supervised Machine Learning Models for Fraud Transaction Detection A Big Data Perspective in Finance can be described as a very topical and successfully conducted research on credit card fraud detection. It proposes a strong methodology based on SMOTE method of data balancing, it uses a real-world dataset and it has mainly focused on the problems of imbalanced datasets. The unique value addition is a novel approach to structured financial transaction data with a Convolutional Neural Network (CNN). This method has shown better results with impressive statistics 94.50 percent accuracy and 99.06 percent recall and 1.00 ROC-AUC, which is surprisingly better than classical machine learning models.

The manuscript is structured, clearly written and carried out with a thorough literature